Hamidreza Moaddeli

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Moaddeli • @hrmoaddeli

Research Interests

Data Science, Big Data, Deep Learning, Machine Learning, Parallel Computing, HPC and Scientific Programming

Education

- **Shiraz University** — **Shiraz, Iran**
  M.Sc. in Computer Engineering, Software Engineering Major, GPA: 17.67/20.0 — *2005–2008*
  Master thesis: *"Design and Implementation of a Prototype for Resource Reservation Mechanism in Grid Environment"*
  supervisor: Dr. GH.Dastghaiby Fard
- **Shiraz University** — **Shiraz, Iran**
  B.Sc. in Computer Engineering, Software Engineering Major, GPA: 17.66/20.0 — *2000–2005*
  Final year project: *"Design and Implementation of NACHOS in Python Programming Language"*
  supervisor: Dr. A. Towhidi

Honors and Awards

- Ranked 24th among more than 400,000 participants in Iranian National University Entrance Exam (B.Sc.), 2000.
- Singapore International Graduate Award (SINGA), A*STAR, Singapore, January 2013.
- Ranked 1st place in B.Sc., Computer Science and Engineering Department at Shiraz University, 2005.
- Ranked 78th in Iranian National M.Sc. University Entrance Exam (Software Engineering Track), 2005.
- Ranked 31st in Iranian National M.Sc. University Entrance Exam (Artificial Intelligence Track), 2004.
- Ranked 52nd in Iranian National M.Sc. University Entrance Exam (Computer Architecture Track), 2004.
- 4-year Exceptional Talent Undergraduate Student Award (Academic-based), 2000-2004.
- Member of Iranian Elite Foundation, 2008-present.

Selected Courses

Big Data Analysis with Apache Spark: edX (Online Course), September 2016 — *Verified Certificate: 100/100*
Introduction to Apache Spark: edX (Online Course), September 2016 — *Verified Certificate: 100/100*
Distributed Machine Learning with Apache Spark: edX , August 2016 — *Verified Certificate: 100/100*
Big Data(a series of two courses): BerkeleyX-edX, August 2015 — *XSeries Certificate*
Scalable Machine Learning: edX (Online Course), August 2015 — *Verified Certificate: 100/100*
Introduction to Big Data with Apache Spark: edX (Online Course), July 2015 — *Verified Certificate: 100/100*
Introduction to R: DataCamp (Online Course), June 2015
Design and Analysis of Algorithms: Coursera (Online Course), Spring 2012, Stanford University — *100/100*
Machine Learning: Coursera (Online Course), Fall 2011, Stanford University — *400/400*
Artificial Intelligence: Coursera (Online Course), Fall 2011, Stanford University — *97.8/100(top 10%)*

Technical and Personal skills

General Skills: Excellent communication and presentation skills, Problem solving skills with strong background in mathematics, Collaboration and teamwork skills, Self-motivated learner.

Programming Languages: Python, C/C++, Octave, R, Java, JavaScript, Angular, HTML

Data Science and Machine Learning: Python SciPy Stack (SciPy, NumPy, Matplotlib, Pandas, scikit-learn), Spark, TensorFlow, Keras, PyTorch

Programming Tools: CUDA, MPI, OpenMP, Git

Systems: Windows, Linux/Unix, Embedded Linux systems, and Container(Docker)

Language Proficiency: Fluent in English, Native in persian

Professional Experience

Work Experience

- **Freelance** — **Shiraz,Iran**
  Freelance Python Developer/Data Scientist — *Sept.2014–Present*
  - Developing softwares for scientific computing and data analysis using *SciPy* Stack (*SciPy, NumPy, Matplotlib, Pandas* and *IPython Notebook.)* and *Mysql* and *SQLite* DataBases.
- **Pars Hassas Co.** — **Shiraz,Iran**
  Part-time Automation Software Engineer/Consultant — *Sept.2014–Present*
- **Pars Hassas Co.** — **Shiraz,Iran**
  Senior Automation Software Engineer — *Dec.2010–Sept.2014*
  - Designing and implementing HMI softwares, logic programs, loop controls and network configuration for multiple small to large projects including Yokogawa DCSs, Siemens PLCs and HIMA ESDs.
  - Engineering Designing and preparing proposal documents for multiple small to large Oil,Gas and petrochemical industries bids based on Yokogawa DCS, ABB DCS, Yokogawa SIS, HIMA ESD.
  - Worked as a main developer for the project *"Design and implementation a SCADA and Telemetry system for Southern water supply network of Shiraz city"*
- **Pars Hassas Co.** — **Shiraz,Iran**
  Automation Software Engineer — *Sept.2008–Dec.2010*
  - Worked as a main developer for the project *"Design and development of a new monitoring and control system for PSA Unit of Bandar-Abass Oil Refinery"*.
- **Pars Hassas Co.** — **Shiraz,Iran**
  Part-time Automation Software Programmer — *Sept.2005–Sept.2008*
  - Developed Industrial monitoring and control softwares using Visual Basic and C++.

Teaching Experience

- **Azad University of Shiraz** — **Shiraz,Iran**
  Lecturer — *2009–2010*
  Courses: *Assembly Language Programming, Discrete Mathematics, Data Storage and Retrieval* and *Artificial Intelligence*
- **Faculty of Engineering Shahid Bahonar Shiraz** — **Shiraz,Iran**
  Lecturer — *Fall 2005*
  Courses: *Computer Graphics Lab* and *Fundamentals of Web*
- **Shiraz University** — **Shiraz,Iran**
  Teaching Assistant — *2003–2006*
  Courses: *Fundamentals Of Programming (Python Programming), Discrete Mathematics, File Structures, Artificial Intelligence, Design and Analysis of Algorithms, Operating Systems Lab, Operating Systems* and *Advanced Mathematics in Computer Engineering*

Research Experience

- **Shiraz University** — **Shiraz,Iran**
  Part-time Research Assistant, Advisor: Dr. F. Khunjush — *Sept.2009–May.2012*
  Research focused on:
  - Studying different performance analysis and modeling methods for GPU Architectures.
  - Using performance counters to develop a power usage model for GPU Architectures.
  - Developing an accurate GPU performance model based on using performance counters and different profiling tools such as CUDA profiler and CUPTI API.
  - Exploring the memory hierarchy in the Fermi GPU Architecture and measuring various memory characteristics.
  - Employing Machine learning and Statistical techniques in Performance optimization and modeling (specifically for many-core architectures).

Publications and Presentations

- **H. Moaddeli**, G. Dastghaibyfard, and M. Moosavi, "Flexible advance reservation impact on backfilling scheduling strategies," *in Grid and Cooperative Computing, 2008. GCC 08. Seventh International Conference on, Oct 2008*, pp. 151–159.
- **H. Moaddeli**, "Practical Deep Learning in Python", Workshop, The third international conference on Pattern Recognition and Image Analysis (IPRIA2017) http://www.ipria2017.com/ Shahrekord, Iran, April 2017. (Workshop)
- **H. Moaddeli**, "GPU Profiling", presented at GPU Workshop, IPM (The Institute for Research in Fundamental Sciences) HPC Laboratory http://hpc.ipm.ac.ir/ Tehran, Iran, Oct 2011. (Invited Talk)